1.4 Exact ODEs. Integrating Factors

1.5 Linear ODEs. Bernoulli Equation. Population Dynamics

1.4 Exact ODEs. Integrating Factors

given

$$\boxed{M\,dx + N\,dy = 0}$$

Aim: find $u(x,y)$ s.t.

$$\frac{\partial u}{\partial x} = M \quad \& \quad \frac{\partial u}{\partial y} = N$$

Solution:

$$u(x,y) = \text{constant}$$

Example

$M\,dx + N\,dy = 0$

$$\underbrace{(1 + 2xy^3)}_{M}\,dx + \underbrace{3x^2y^2}_{N}\,dy = 0 \qquad (*)$$

$\frac{\partial u}{\partial x} = M$ & $\frac{\partial u}{\partial y} = N$

$$u = x + x^2y^3$$

solution y for $(*)$ can be solved from

$x + x^2y^3 = \text{constant}$

$u(x,y) = C$

differential of both side

$$\frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy = 0$$

ODE becomes

$\rightarrow du = 0$

$\boxed{u = c}$

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy$$

Not all equation of the form $M\,dx + N\,dy = 0$ are exact.

Only those which satisfy

$$\boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$$

why?

EXAMPLE 1

then solve

$$\underbrace{\cos(x+y)}_{M}\,dx + \underbrace{(3y^2 + 2y + \cos(x+y))}_{N}\,dy = 0$$

Q: find $u(x,y)$ s.t. $\frac{\partial u}{\partial x} = M$ & $\frac{\partial u}{\partial y} = N$

$$\frac{\partial u}{\partial x} = \cos(x+y)$$

$$u = \sin(x+y) + f(y)$$

needs to be solved

$$\frac{\partial u}{\partial y} = N$$

$$\cancel{\cos(x+y)} + \frac{df}{dy} = 3y^2 + 2y + \cancel{\cos(x+y)}$$

$$f(y) = y^3 + y^2 + c$$

$$u(x,y) = \sin(x+y) + y^2 + y^2 + C$$

Solution of given ODE is y with satisfies the equation

$$\boxed{\sin(x+y) + y^2 + y^2 = C}$$

Example: Check for exactness

$$-y\,dx + x\,dy = 0$$

$$-y\,dx + x\,dy = 0$$

$$(e^{x+y} + ye^{y})\,dx + (xe^{y} - 1)\,dy = 0, \quad y(0) = -1$$

1.5 Linear ODEs. Bernoulli Equation. Population Dynamics

Linear 1st order ODE

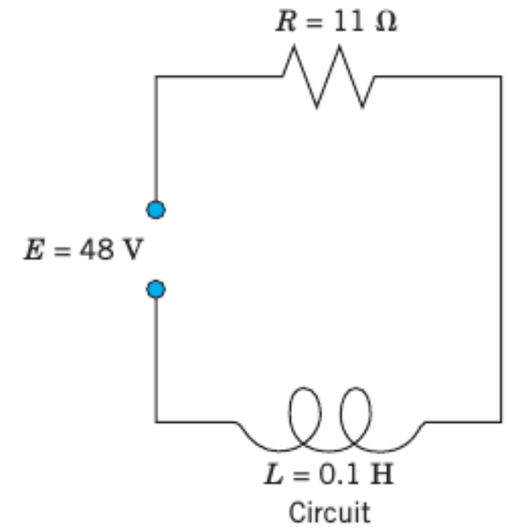
$$y' + p(x)y = r(x)$$

Solve the initial value problem

$$y' + y \tan x = \sin 2x, \qquad y(0) = 1.$$

EXAMPLE **Electric Circuit**

$$LI' + RI = E(t)$$



$E = 48$ V

$R = 11\ \Omega$

$L = 0.1$ H

EXAMPLE 3 **Hormone Level**

$$y'(t) = \text{In} - \text{Out}$$

$$y'(t) = A + B\cos\omega t - Ky(t)$$

Reduction to Linear Form. Bernoulli Equation

$$y' + p(x)y = g(x)y^a$$

EXAMPLE 4 **Logistic Equation**

$$y' = Ay - By^2$$